HeatRelief
Judge-Winning Upgrade Plan

Full audit of the existing project, the detailed Action Plan page (v2) specification, AWS add-on services, and a step-by-step handoff for Claude Code.

Event: Environmental Hacks (WeMakeDevs x AWS, Bharat Builds Tour), Heat and Water track, 8 to 11 Oct 2026.
Prepared on 10 Oct 2026 from your project documentation, 7 screenshots and the event page.

Part	What it contains
0. Read first	Where you stand against the five judging criteria, triage tiers (T0 to T3), what NOT to touch, and a time-boxed order of work.
1. Action Plan page v2	Detailed spec: flow, wireframe, every block with its logic and data source, the flaws it fixes, API additions, water layer, presets, acceptance tests.
2. Weakness register	Every flaw found (documentation plus what is visible in your screenshots), the fix written as an instruction, and a 'done when' test.
3. AWS add-ons	Services to add on top of your existing 14, what a judge actually sees, effort, cost cautions, services to avoid, and a 3-minute video storyboard.
4. Claude Code handoff	Ground rules, ordered task list with acceptance tests, and the final submission checklist.

How to read the tiers. T0 = delivery, must do or you risk disqualification or an unverifiable project. **T1** = judge-visible, do tonight. **T2** = do if time remains after T1. **T3** = roadmap only: mention it honestly, do not build it. Estimates are rough working hours for one person with Claude Code. Items marked 'verify' depend on facts I could not confirm from here (prices, service limits, official criteria): check the current source before relying on them.

Part 0. Read first

0.1 What the judges score, and where HeatRelief stands

The event page lists five criteria. The project is strong on engineering and honesty, but several things a judge can verify are missing or not yet proven. Your **biggest risk is not a missing feature: the AWS stack has never been deployed** and the Bedrock path has only run against a mock (your own checklist, section 11.3).

Criterion	What a judge sees today	Gap	What moves it	Tier
Idea and impact	Clear decision chain (where, who, what to build, who to warn). Honest limits. Real Landsat data for 4 cities.	The 'Water' half of Heat and Water is almost absent. Costs and cooling are placeholders. Nothing shows what a plan changes versus a naive plan.	Water layer (W1 to W3), naive-plan benchmark number, '% of hotspot cells treated', equity-versus-efficiency headline sentence.	T1
Built on AWS (mandatory for a prize)	14 services designed in one SAM template. Strands Agents and SAM CLI are used (both AWS open source).	Not deployed. Bedrock untested live. Nothing a judge can click. The rules say a project must use an AWS open-source tool or be deployed on AWS.	Deploy, live Bedrock brief with an on-screen 'Generated by Amazon Bedrock' badge, console cuts in the video, Cognito and Step Functions add-ons.	T0
Design and usability	Clean, consistent UI, light and dark, Hindi toggle, mobile citizen page.	Visible defects in your screenshots: raw floats, overlapping tiles, 'wards' versus zones, flat chart, identical compare columns, 5 km 'nearest' cooling places.	Fix list B1 to B9 (Part 2). All are small and all are visible.	T1
Execution	53 tests pass, golden-path smoke test, graceful fallbacks.	Cold start with a 29 s limit, a global 10 requests per second throttle, unauthenticated plan approval.	Provisioned concurrency for the judging window, higher throttle, Cognito on write routes.	T0
Demo video (3 min, no live demo)	Not mentioned anywhere in the docs: treat as not started.	Today's alert is GREEN (33.2 C), so the alert, SMS-style email and Hindi voice story would be invisible on camera.	'Simulate heatwave' toggle (B5) and the storyboard in Part 3.	T0

Other event facts that affect you (from the event page): the hackathon runs online Thursday 8 Oct to Sunday 11 Oct. Entrants must verify as a student on AWS Builder Center. The top 5 blog posts published on AWS Builder Center and linked in the submission win AirPods. Top 4 runners-up get AWS credits. Confirm the exact submission deadline time on the event page or Discord tonight; I could not find a clock time.

0.2 Triage: what to do, in what order

ID	Item	Why a judge sees it	Est.	Tier
A1	Deploy the SAM stack to AWS and fix what breaks	A live URL is the proof for 'Built on AWS' and 'Execution'	2 to 3 h	T0
A2	Enable a Bedrock model, test briefs live (English and Hindi), test the guardrail in Hindi	'Generated by Amazon Bedrock' badge, real agent output	1 to 1.5 h	T0
A3	Warm-up: provisioned concurrency 1, raise API throttle, precompute heavy results	No cold-start spinner or 429 errors during judging	0.5 h	T0
A4 to A6	Repo history linked, README, student verification, team form, confirm deadline	Eligibility	0.5 h	T0
V1	3-minute demo video using the storyboard	It is the only thing judges watch	1.5 h	T0
B1 to B5, B9	Number formatting, zone naming, compare block, flat chart, simulate-heatwave toggle, city grouping	Visible polish and the alert story	2 h	T1
P1	Action Plan v2 core blocks R1 to R6, R8, R9, R11 (CSV and GeoJSON)	The page judges will spend most time on	4 to 5 h	T1
W1 to W3	Water points, tree irrigation demand, tanker priority list	Makes it a Heat AND Water project	3 to 4 h	T1
S1, S2	Cognito for officer actions, Step Functions plan lifecycle with SNS fan-out per department	Visible AWS depth in the video	3 to 4 h	T1
S3	Tracing and a Bedrock-versus-template metric on the CloudWatch dashboard	Cheap, shows in console cut	0.5 h	T1
BL	Blog post on AWS Builder Center	Separate prize, also a submission link	1 h	T1

ID	Item	Why a judge sees it	Est.	Tier
C3, E2, F1, F2, R7, R10 to R14	Rank stability, public/private tag, IMD-aligned thresholds, Kannada/Marathi text, sensitivity curve, verification plan, citizen boost, plan PDF	Credibility and operations	1 to 2 h each	T2
Roadmap	Night-time layer, official ward polygons with census age data, persistent hotspots, real roof areas, post-implementation measurement	Say them honestly in the video and README, do not build	-	T3

Minimum winning set if you only have about 8 hours: A1, A2, A3, A4 to A6, B1 to B5, P1 (blocks R1 to R6, R8, R9), W1 and W3, V1. Everything else is a bonus. Do not start a T2 item until every T0 and T1 line above is ticked.

0.3 Do NOT do (protects your time)

- **Do not rebuild or restyle** existing pages, rename API routes, or change the model. Everything below is additive.
- **Do not add the 'use today's alert level' multiplier to the plan optimizer** (suggested in an earlier draft). Your alert multiplier (1.0 to 2.0) is applied uniformly to every ward, and a uniform multiplier cannot change a ranking or a greedy selection. It would be cosmetic. Use alert level to drive the *response actions* (SOP, tanker list) instead.
- **Do not add** OpenSearch Serverless, Pinpoint, EKS, Fargate or SageMaker endpoints (reasons in Part 3.3).
- **Do not invent citations or numbers.** Costs and cooling stay labelled as planning-grade placeholders unless you can verify a source tonight.
- **Do not claim wards.** Your zones are 2 km blocks unless official ward polygons are loaded.

Part 1. Action Plan page v2: detailed specification

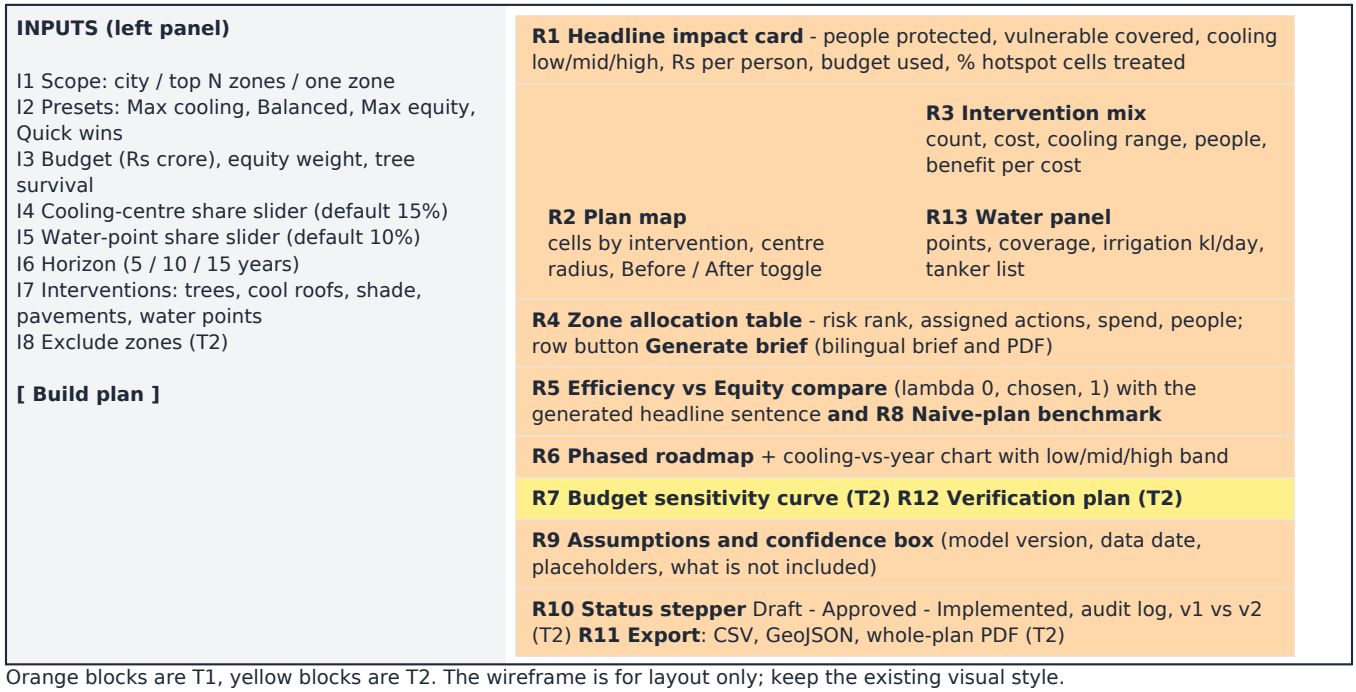
1.1 What is wrong with the page today

From your screenshot the page shows one row of four numbers, a bullet list of caveats, an unlabelled status chip, two buttons, an efficiency-versus-equity box, and a chart. The backend already returns far more than the UI shows (impact card, timeline, per-item cost, people covered, low/mid/high cooling, compare, status with audit, exports).

#	Observed problem (screenshot or doc)	Why it hurts
1	No map. The plan is a list of numbers; you cannot see where anything goes.	Judges grasp a map in two seconds. Biggest single visual gap.
2	Efficiency plan and Equity plan boxes look almost identical: 'Spend in high-vulnerability cells: 100%' on both, average vulnerability 97% versus 98%, cooling -1.2 on both.	The compare feature is your best argument, but the rows shown carry no information. The real difference is in people protected: 2,045,959 versus 2,880,414, i.e. 834,455 more people (about 41% more) under the equity plan.
3	Cooling-over-time chart renders as a flat line at 0 in the screenshot.	Looks broken. Even if only cut off, the low/mid/high band is not visible.
4	Cooling shown as one number per box in the compare and as '-1.9 to -0.5 C' in the headline only.	Honesty rule says ranges everywhere.
5	Headline says Rs 49.8 cr spent but never says how much of the city was treated.	A few dozen cells out of roughly 10,000 reads as underwhelming. Show '% of hotspot cells treated'.
6	Hidden inputs: cooling-centre share (15%), horizon (10 years) and scope are fixed in code.	Officers cannot express 'only this zone' or 'quick wins first'.
7	No link to Ward Risk: the 'Plan for this ward' button has nowhere specific to land, and the brief/PDF is not reachable from the plan.	Breaks the decision chain you describe.
8	Plan status chip and Approve button exist, but audit log, versions and role control are not shown.	A governance feature that is invisible earns no credit.
9	Units: UI says Rs crore, backend works in lakh.	1 crore = 100 lakh. A silent factor-of-100 error is the worst possible bug in front of judges. Add a unit test.

1.2 Page flow

Restructure as a five-step flow with a sticky step bar, not one long scroll: **Inputs, Plan result, Compare, Review and approve, Brief and export.** Steps 1 and 2 are the left and right panels; steps 3 to 5 are tabs or collapsible sections under the result so the page fits a 3-minute video.



Orange blocks are T1, yellow blocks are T2. The wireframe is for layout only; keep the existing visual style.

1.3 Block-by-block specification

Each block lists what it shows, where the data comes from (existing = already in the API per your documentation; new = small addition), the logic, and the flaw it fixes.

Block	What it shows	Data / logic	Flaw fixed	Tier
I1 Scope	Whole city, top N zones by risk rank, or one zone. Pre-filled when arriving from Ward Risk via 'Plan for this ward' (query param zone_id).	New param `scope`. Filter the cell set before the optimizer runs. Zone brief and plan then use the same cell set.	Plan disconnected from ward ranking	T1
I2 Presets	Four buttons that fill the controls (table 1.4). Controls stay editable afterwards.	Front-end only; sends the resulting params.	Officers cannot express intent quickly	T1
I4 / I5 / I6	Sliders for cooling-centre share (0 to 30%, default 15), water-point share (0 to 20%, default 10), horizon 5/10/15 years.	Existing optimizer constants become request parameters with the current defaults, so old calls behave identically.	Hidden inputs	T1
R1 Headline card	People protected; **vulnerable people covered** (new); cooling as low / mid / high; Rs per person protected; budget used of budget; **hotspot cells treated = n of N (x%)**.	Existing impact card + new fields `vulnerable_people_covered` (population in treated cells and centre catchments whose cell vulnerability percentile is 0.8 or more), `hotspot_cells_total`, `hotspot_cells_treated` (hotspot = critical or high tier).	Single-number cooling; 'few cells' looks small	T1
R2 Plan map	Same Deck.gl map as Heat Map. Treated cells coloured by intervention (5 colours plus legend). Cooling centres and water points as icons with a circle at the walking radius used in the optimizer. **Before / After toggle**: After = predicted LST minus mid cooling, shown in tier colours.	Plan response already lists cells with intervention and cooling; add lat/lon (or cell id and join to the grid you already load).	No map	T1
R3 Intervention mix	Per intervention: cells, cost (Rs lakh), cooling low to high, people covered, benefit per lakh, years to full effect.	Group the plan items by type. Sort by benefit per lakh.	Cost transparency	T1
R4 Zone allocation	Per zone: risk rank, actions assigned, spend, people covered, (T2) citizen reports supporting. Row button **Generate brief**.	Group plan cells by zone id. Button calls the existing `POST /brief` with zone and the plan budget, opens the bilingual brief and PDF in a side drawer.	Plan disconnected from the brief	T1
R5 Compare	Three columns: lambda 0 (efficiency), your chosen lambda, lambda 1 (equity). Rows: people protected, vulnerable covered, cooling mid with range, cells treated, spend. Headline sentence generated by code, never by an LLM: 'The efficiency plan misses N vulnerable people that the equity plan covers.'	Existing compare option, extended to return the chosen-lambda column. N = equity.vulnerable_covered minus efficiency.vulnerable_covered, shown only if positive; otherwise show the neutral sentence.	Identical-looking boxes	T1
R8 Naive benchmark	'Optimizer versus hottest-cells-first at the same budget': people protected, vulnerable covered, people per Rs crore, and the multiple (for example 'covers 3.1x more people per rupee').	New function (section 1.5). Baseline definition is printed under the numbers so it is honest.	Shows the optimizer's value in one number	T1
R6 Phased roadmap + chart	Phase cards: **Now** (cool roofs, shade, cooling centres, water points: 0 years), **Year 1** (pavements), **Years 1 to 8** (tree ramp). Under it a line chart of cooling versus year with a shaded low-to-high band.	Existing timeline array (year, low, mid, high). Phase membership from each intervention's years-to-full-effect. Y axis must show negative degrees; if the series is all zero show a message instead of a flat line.	Flat chart; no sequencing	T1
R13 Water panel	Water points placed and people with a water point inside walking distance; tree irrigation demand (kilolitres per day); top tanker-priority zones.	Section 1.6.	Heat and Water gap	T1

Block	What it shows	Data / logic	Flaw fixed	Tier
R9 Assumptions box	Always visible. Model version, data date, tree survival used, 'cooling from published ranges', 'costs are planning-grade placeholders', 'treated cell = a bundle of actions inside a 500 m cell, not the whole cell', 'daytime surface temperature only, no night layer', 'no age, income or informal-settlement data', 'zones are 2 km analysis blocks'.	Model version and data date from the existing X-Model-Version and X-Data-Date headers; flags from the existing method.not_included field.	Honesty on the page	T1
R10 Status stepper	Draft - Approved - Implemented stepper, audit table (time, actor, note), Approve and Revert buttons enabled only for the officer role.	Existing status and audit; Cognito group check (S1). T2: save as v2 and show v1 versus v2 deltas.	Invisible governance	T1
R11 Export	CSV and GeoJSON (exist). T2: whole-plan one-page PDF reusing the fpdf2 brief code.	Add a plan-summary PDF function beside brief_pdf.	Only single-zone PDF today	T1
R7 Sensitivity curve	People protected versus budget (Rs 10 to 100 cr), with the chosen budget marked and the 'knee' where extra spend stops helping.	Run the optimizer at 10 budgets. Precompute and cache (A3); never compute on request (29 s limit).	Shows diminishing returns	T2
R12 Verification plan	Per action: how success is measured (re-run the Landsat composite next hot season and check change in LST of treated versus matched untreated cells), owner and planting season.	Static template text. Owner and season must carry the label 'template guidance, not computed from your data'.	'How will we know it worked?'	T2
R14 Citizen boost	'N citizen reports support this action' per zone.	Existing /reports/summary; boost in section 1.7.	Links citizen page to plan	T2

1.4 Preset definitions

Preset	Equity weight lambda	Interventions allowed	Other settings	Meaning
Max cooling	0.0	all	-	Most estimated cooling per rupee in the hottest, most suitable cells
Balanced	0.5	all	-	Your current default behaviour
Max equity	1.0	all	-	Cooling delivered to vulnerable people first
Quick wins	0.5	only those with 0 years to full effect: cool roofs, shade structures (plus cooling centres and water points)	horizon 5 years	Everything that helps this season; no trees, no pavements

1.5 Core logic you must implement exactly

Units. Internal unit is Rs lakh. 1 crore = 100 lakh. Put one helper `lakh_to_crore(x) = x / 100` in the backend and one in the frontend, and add a test: a Rs 50 cr budget must equal 5,000 lakh in the optimizer.

Hotspot share. hotspot cells = cells in the critical and high tiers (tiers already exist). Show treated hotspot cells divided by all hotspot cells, as a percentage with the counts.

Naive baseline (R8). Defined and printed on the page as 'hottest cells first, strongest intervention, no cost or equity ranking':

```
def naive_plan(cells, budget_lakh, allowed):
    plan, spent = [], 0.0
    for c in sorted(cells, key=lambda c: -c.lst):            # hottest first
        feas = [i for i in allowed if c.suit[i] > 0]
        if not feas: continue
        iv = max(feas, key=lambda i: abs(COOL[i].mid))        # strongest cooling, ignores cost
        if spent + COST[iv] <= budget_lakh:
            plan.append((c, iv)); spent += COST[iv]
    return metrics(plan)   # same people_protected / vulnerable_covered code as the optimizer
# ratio shown = optimizer.people_protected / naive.people_protected  (same budget)
```

Compare headline (R5). Compute deterministic text from numbers, for example: `f"The efficiency plan misses {n:,} vulnerable people that the equity plan covers."` Only show when n is above 0.

Roadmap (R6). Group the plan's interventions by `years_to_full_effect`: 0 = Now, 1 = Year 1, 8 = Years 1 to 8 (ramp). Cost per phase = sum of item costs. The cooling chart uses the existing timeline array.

Sensitivity (R7). Budgets 10,20,...,100 crore, same lambda and interventions. Knee = first budget where the marginal people per extra crore falls below 25% of the first step. Store the series in S3 as `plan_sensitivity/<city>.json` at build time and serve it from a GET route.

1.6 Water layer (what makes it Heat AND Water)

Your docs use water fraction as a feature (it is the top SHAP driver at 1.23, and MNDWI is fourth at 0.57), yet no intervention, alert or plan output touches water. Add three small things that reuse existing code paths. Every figure below is a configurable placeholder, labelled as such on screen.

ID	Feature	How it works	Where it appears
W1	Drinking-water points as an intervention	New type `water_point`: no cooling, cost from config (placeholder lakh per point). Placed exactly like cooling centres: greedy maximise vulnerable people not yet covered within walking distance, from its own share of budget (default 10%). It does not consume a cell's single intervention slot.	Mix table, map icons, headline 'people with a water point nearby'
W2	Water cost of the plan	water_kl_per_day = trees_planted x litres_per_tree_per_day / 1000 x survival. trees_per_cell and litres_per_tree_per_day are config constants named PLACEHOLDER_*. Also show the guidance line 'prefer low-water native species; irrigate from treated wastewater where available' as text only.	Water panel (R13)
W3	Heat-day tanker and kiosk priority list	For zones: score = risk x (1 - water_point_coverage_share) x (1 + 0.25 x min(no_water_reports, 4) / 4). Your citizen page already collects `no_water` reports; this uses them. Top 10 zones shown. On orange or red days the daily Lambda publishes this list to a Water Board SNS topic (S2).	Water panel, daily alert email
W4	Waterbody revival, treated-wastewater irrigation	Roadmap text only. Do not build.	README and video closing

Story for judges: 'Our model's two strongest drivers are water-related (SHAP is explanatory, not causal). So we plan for water twice: places to drink and places to stay cool, and the water the trees will need.' State clearly that costs, litres and trees per cell are placeholders.

1.7 Citizen report loop (T2)

Boost a cell's benefit by `(1 + 0.25 x min(reports_in_cell, 4) / 4)` where reports are `no_shade` or `very_hot` in the last 90 days. Show 'N citizen reports support this action' per zone. Spam control: one report per device token per cell per day (store a hashed token, not a location), API Gateway route-level throttle on `POST /reports`, and Rekognition moderation for photos (S5). Reports are anonymous cell ids already; keep it that way.

1.8 API additions (additive, old calls unchanged)

```
POST /api/v1/plans/optimize
request adds:  scope:{type:"city|top_n|zone", n:10, zone_id:"Zone 20-10"}, centre_share:0.15, water_share:0.10,
               horizon_years:10, exclude_zones:[], include_benchmark:true
response adds:
 impact: { ..existing.., vulnerable_people_covered, hotspot_cells_total, hotspot_cells_treated,
           cooling:{low,mid,high} }
 mix:    [{type, cells, cost_lakh, cooling:{low,mid,high}, people, benefit_per_lakh, years_to_full}]
 zones:  [{zone, risk_rank, assigned:{trees:3, cool_roofs:1}, cost_lakh, people, reports}]
 compare:{ lambda0:{..}, chosen:{..}, lambda1:{..}, headline:"..." }
 benchmark:{ naive:{people,vulnerable}, optimizer:{people,vulnerable}, people_per_crore_ratio }
 roadmap:[{phase:"Now", items:[..], cost_lakh}], timeline:[{year, low, mid, high}]
 water:  { points, coverage_people, irrigation_kl_per_day, tanker_priority:[{zone,score}], assumptions:{..} }
 assumptions:{ model_version, data_date, flags:[..] }
GET  /api/v1/plans/sensitivity?city=new_delhi&lambda=0.5      -> precomputed series from S3
```

1.9 Acceptance checklist for the page

- A Rs 50 cr plan on Delhi shows 'cells treated: n of N hotspot cells (x%)' and a cooling range, never a single cooling number.
- The map shows treated cells and a Before/After toggle that changes colours; centres show a walking-radius circle.
- R5 prints a generated sentence with a real difference in vulnerable people, or the neutral sentence if there is none.
- R8 prints the baseline definition and a multiple; ratio equals optimizer people divided by naive people at the same budget.
- The chart is never a flat zero line: it shows a band or a clear message.
- 'Generate brief' on a zone row opens the English and Hindi brief and the PDF.

- Quick wins preset produces a plan with zero trees and zero pavements.
- Unit test: Rs 50 cr = 5,000 lakh. Existing 53 tests still pass. New tests for naive_plan, hotspot share, water coverage.

Part 2. Project-wide weakness register and how each is fixed

Evidence is tagged **[doc]** (your documentation) or **[shot]** (visible in a screenshot). The fix column is written as an instruction for Claude Code. 'Done when' is a test you can run in two minutes.

A. Delivery and AWS proof

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
A1	Stack never deployed; nothing for a judge to click. [doc 9, 11.3]	Run the SAM deploy script with credentials, Docker and SAM CLI. Fix build errors one by one; set the alert email and Bedrock model id parameters. After deploy run the smoke test against the live API URL. Write the two stack URLs into the README.	Public CloudFront URL loads the Delhi map; /health OK; golden-path smoke passes against the live URL; budget alarm active.	T0
A2	Bedrock only tested against a mock; Hindi quality unchecked. [doc 11.3]	Enable model access in the Bedrock console for 2 or 3 candidate models. Run 10 briefs each; pick the model with the most `source=bedrock` and `numbers_verified=true` and good Hindi (ask a Hindi reader). Test the guardrail with a Hindi prompt: guardrail language coverage may differ from English, so confirm it neither blocks valid briefs nor passes off-topic ones (verify). Narrow the Bedrock IAM statement to the chosen model or profile ARN and the guardrail ARN. Add a visible badge 'Generated by Amazon Bedrock, numbers verified' or 'Template fallback' from the response's source field.	At least 8 of 10 briefs return source=bedrock with numbers_verified true; badge visible on the UI; IAM no longer uses wildcard for Bedrock.	T0
A3	29 s API Gateway limit plus Lambda cold start that loads data and model; global throttle of 10 requests per second with burst 20 can return 429 when several judges and the map's parallel calls arrive. [doc 9, 10]	Set provisioned concurrency 1 on ApiFunction for the judging window (small daily cost, verify current price). Raise throttle to about 50 per second, burst 100, and keep a lower throttle only on POST /reports. Precompute slow things (sensitivity, rank stability) into S3 at build time. Keep /plans/optimize with compare under about 10 s; log duration. Warm with /health then the ward risk call before recording.	Cold first request under 10 s; 30 parallel requests produce no 429.	T0
A4	Git history re-initialised; original history unlinked. [doc 11.3]	Link or push the original history to the public repo before submission; keep the commit log honest. Add a README with architecture diagram, honesty statement and live URLs.	Repo public; README has URLs and diagram.	T0
A5	No demo video or blog in the docs. Video is the only thing judges watch.	Record with the storyboard in Part 3.4. Publish the blog on AWS Builder Center (problem, stack, what fought back) and link it.	Video under 3:00; blog link in submission.	T0
A6	Eligibility steps unconfirmed.	Verify student status on AWS Builder Center, check in on the event page, team leader fills the credits form if needed, confirm the submission deadline.	All four ticked.	T0

B. UI defects visible in your screenshots

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
B1	Ward Risk shows raw floats ('46.54600143432617 C'), and the Avg surface temp tile overflows into the People tile. [shot]	Format temperatures to 1 decimal in the tiles and the table. Use tabular numbers, set `min-width:0; overflow:hidden; text-overflow:ellipsis` on tiles, and thousands separators for people.	No tile overlap at 1280 px and at mobile width; all temperatures show 1 decimal.	T1
B2	Title 'Which wards need help first?', nav 'Ward Risk' and 'Ward / zone' while zones are 2 km blocks (Zone 20-10, 624 of them). [shot, doc 3]	Rename visible text to 'analysis zones'. Read the existing method field (`2km_analysis_zones` versus `official_boundaries`) and show 'Ward' only when official boundaries are loaded. Apply in English and Hindi strings.	Search the UI for 'ward': it appears only when official boundaries are active.	T1
B3	Compare boxes look identical. [shot]	Replace with block R5 (Part 1).	See 1.9.	T1
B4	Cooling chart flat at 0. [shot]	Block R6: band, negative axis, empty-state message.	See 1.9.	T1

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
B5	Alert is GREEN (33.2 C) in October, so alert, email and voice features are invisible in a demo. [shot]	Add `?simulate=yellow|orange|red` to /risk/today and /citizen. Response carries `simulated:true`; UI shows a striped SIMULATION banner. Allow the daily Lambda to be invoked manually with `{"simulate":"red"}`; the SNS subject gets the prefix [SIMULATION]. Never present simulated output as real.	Opening /citizen?simulate=red shows a red card and banner; manual invoke sends the email and Hindi audio.	T1
B6	Citizen page: nearest 'places to cool down' are 4.8 to 5.3 km away, one is just called 'Park'; rank and zone are shown before location consent. [shot]	Show walking minutes. If nothing is within 1.5 km, say so plainly and show the nearest water point. Hide the rank line until location is shared, or label it 'for the default zone'. Skip generic names or show type only. T2: Amazon Location Service walking route (S6).	No place beyond 1.5 km is described as 'nearest' without a warning.	T2
B7	Scenarios page: Run button disabled at 0%, Before and After maps drawn at different sizes, and its interventions (green roofs, water bodies, albedo) differ from the Plan page (trees, roofs, shade, pavements). [shot]	Sync the two map view states. Add a one-line note stating the Scenarios page is an exploratory what-if and the Plan page is the budgeted tool. T2: button 'Use these in a plan'. If time is short, leave it out of the video.	Both maps share zoom and centre.	T2
B8	Heat Drivers: every bar is the same red, so direction is hidden; text mentions building density though footprints are off; 'Water' ranks first with no explanation. [shot, doc 3]	Colour bars by direction (sign of the correlation between feature value and SHAP value; heating red, cooling blue). Remove the building-density mention unless footprints are enabled. Add: 'Water-related features are the strongest drivers in our model; SHAP explains the model, not causation.'	Bars show two colours; text matches enabled features.	T2
B9	City list mixes 4 real cities and 16 synthetic previews. [doc 1]	Group the dropdown: 'Real satellite data' (4 cities) first, 'Synthetic preview' below, default New Delhi.	Dropdown has two groups.	T1

C. Data and model credibility

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
C1	Costs and cooling are placeholders; 'where does Rs 120 lakh per cell come from?' [doc 5.4, shot]	Move numbers into one config file with low, mid, high and a `source_note` per row. Show cost as a range in the mix table. Only add a citation if you can verify it tonight; otherwise keep the label 'planning-grade placeholder'. Define 'treated cell' on screen as a bundle of actions inside a 500 m cell (25 ha), not a whole-cell conversion.	No cost shown without 'placeholder' label or a verified source.	T1
C2	City-level bias: Bengaluru +3.7 C, Jaipur -3.2 C on unseen cities. [doc 4]	Already mitigated by showing observed Landsat for tiers. Keep the bias column on the Model Card and state it in the video. Do not build bias correction tonight; roadmap is station air-temperature fit per city.	Model Card shows bias; assumptions box mentions it.	T3
C3	Ranking confidence unknown.	Script `models/rank_stability.py`: 200 runs jittering risk component weights (hazard 0.6/0.4 split, vulnerability weights by plus or minus 0.15) and dropping one component; write the share of runs each zone stays in the top 10. Show 'Top-10 in N% of runs' chip on Ward Risk.	Chip appears for top 10 zones; JSON in S3.	T2
C4	Population from Meta HRSL may undercount dense slums. [doc 3]	Add one line to the assumptions box. Roadmap: audit against Census ward totals.	Line present.	T1
C5	OSM cooling spots partial (Delhi 5,399, Mumbai 2,351, Jaipur 895, Bengaluru 998); a listed place might be closed. [doc 11.3]	Show 'OSM partial' in the data card; add 'unverified' tag to spots; let citizen `cool_spot` reports confirm. Re-fetch tiles with back-off only if time remains.	Tag visible on citizen page.	T2
C6	Night heat missing: nights cause heatwave deaths and the data is daytime Landsat only.	Say it in assumptions and video. Roadmap: add a night surface-temperature source with night passes (check ECOSTRESS or MODIS availability).	Line present.	T3

D. Vulnerability and actionability

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
D1	Vulnerability uses density, green deficit and a crowding proxy; building density is empty because footprints are off. [doc 5.1, 11.3]	Keep. Show the 'not included' line on the plan page too. If an official ward file is available, drop it at `data/external/<city>_wards.geojson`; the pipeline already assigns by centroid (verify licence and CRS). T3: census age structure, informal settlements.	Plan assumptions box lists missing inputs.	T1
D2	No public versus private distinction: a municipality can act on public land directly, private roofs need incentives.	Add an `actionability` column in the mix table from a default mapping: shade structures, cooling centres, water points, pavements = public; trees = public land assumed; cool roofs = mixed. Label it 'default assumption'. Optional toggle 'public targets only'.	Column visible, labelled as assumption.	T2
D3	Real roof area from footprints is off; cool-roof cost is a per-cell guess.	Roadmap only; say so honestly.	-	T3

E. Operations and alerts

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
E1	Alert thresholds are your own (40/42/45 C, heat index 32/41/54) and the multiplier is uniform. [doc 5.2]	Look up the current IMD heatwave criteria (my recollection: for plains, max temperature 40 C or more with a departure from normal of 4.5 C or more, or 45 C or more; verify before using). Align the colour levels and label them 'aligned to IMD criteria' only after checking. Use the level to select an SOP list (E3), not to rescale scores.	Thresholds cite their source in config.	T2
E2	Hindi only; Bengaluru and Mumbai residents use Kannada and Marathi.	Use Amazon Translate on the fixed template sentences (translate once at build, cache); mark 'machine translated'. Polly stays Hindi and English only (verify language list).	Language chips for Kannada and Marathi on the citizen page.	T2
E3	Alerts have no department actions.	Add an SOP table per alert level (which department does what) in config, and publish department-specific messages to separate SNS topics (S2). Optional T2: generate a Heat Action Plan document with pre-season, alert-response and post-season sections, reusing numbers_check.	Red-day email lists department tasks.	T2
E4	No plan check after build.	Add the verification-plan block (R12) and state the roadmap: re-run the Landsat composite after implementation and compare treated cells with matched untreated cells.	Block visible.	T2

F. Security

ID	Weakness and evidence	How we overcome it (instruction)	Done when	Tier
F1	Plan approval, status changes and optimize are unauthenticated. [doc 7]	Add Amazon Cognito: user pool with groups officer and viewer; JWT authorizer on PUT /plans/{id}/status and POST /plans/optimize; read routes and citizen routes stay public. Seed two demo users and show a login chip with the role.	Anonymous Approve returns 401; officer succeeds.	T1
F2	Bedrock IAM uses a wildcard resource. [doc 9.3]	Narrow after the model is chosen (A2).	No Bedrock wildcard.	T1
F3	Citizen report abuse. [doc 7]	Route throttle, one report per hashed device token per cell per day, optional WAF rate rule and Rekognition photo moderation (S5).	Second report same day is rejected.	T2

Part 3. AWS services: what you have and what to add

3.1 Already in your template (14 service types)

Lambda (API and daily refresh), API Gateway HTTP API, EventBridge Scheduler, S3 (data and frontend), CloudFront with Origin Access Control, DynamoDB, Bedrock, Bedrock Guardrails, SNS, Polly, CloudWatch, AWS Budgets, IAM, and CloudFormation through SAM. Strands Agents SDK and SAM CLI are AWS open-source tools. This is a good base: the issue is proof (deployment), not breadth.

3.2 Add-ons ranked by what a judge sees

ID	Service	What it adds in HeatRelief	Judge-visible moment	Effort	Tier
S1	Amazon Cognito (optionally Cedar policies)	Officer and viewer roles; protects Approve and optimize. Cedar is AWS open source if you want a policy file, but Cognito groups are enough.	Login chip with role; 401 for anonymous Approve	2 h	T1
S2	AWS Step Functions plus SNS topics per department (DynamoDB Stream trigger)	State machine 'PlanLifecycle': Draft, Approved (wait for officer), notify Water Board, Health and Municipal Works topics with department-specific text, Implemented, audit write. Same pattern for red-alert SOP fan-out.	Coloured execution graph in the console, three different emails	3 h	T1
S3	CloudWatch custom metrics plus X-Ray tracing (Lambda Powertools)	Metric BriefSource (bedrock versus template), NumbersVerified rate, optimizer duration; active tracing on ApiFunction.	Dashboard widget 'Agent grounded: x%'	0.5 h	T1
S4	Lambda provisioned concurrency	Removes cold start during judging (A3).	No spinner	0.3 h	T0
S5	Amazon Rekognition (moderation and labels on S3 upload event)	Auto-flag unsafe or irrelevant citizen photos; label shade or tree presence as supporting evidence for reports.	Photo report marked 'verified: shade absent'	1.5 h	T2
S6	Amazon Location Service	Walking route and minutes to the nearest cooling place or water point instead of straight-line metres.	Citizen page shows '12 min walk'	2 h	T2
S7	Amazon Translate	Kannada and Marathi text for alerts and citizen advice (E2).	Language chips	1.5 h	T2
S8	Amazon SES	HTML alert email with a table of top zones and the SOP, replacing plain SNS email (sandbox needs verified addresses).	Nicer alert email	1 h	T2
S9	AWS WAF (rate rule on /reports)	Abuse protection (check current price before enabling).	Console screenshot	1 h	T2
S10	Amazon SQS with a dead-letter queue	Buffer citizen reports and async PDF generation.	Little visible value	1 h	T3
S11	Bedrock Knowledge Bases	Retrieval over real heat action plans and guidance instead of one markdown file. Vector store cost can dominate: check it first.	Cited guidance in the brief	3 h	T3

3.3 Services to avoid

- **OpenSearch Serverless** as a vector store: it has a fixed minimum capacity charge that can consume your free credits quickly (verify current minimum before considering).
- **Amazon Pinpoint**: my understanding is that it is ending support around 30 Oct 2026 (verify); use SNS, SES or AWS End User Messaging instead. SMS to India also needs DLT registration, so keep email plus Polly audio.
- **EKS, Fargate, SageMaker endpoints**: more cost and setup, no benefit for this workload. Your model is a small XGBoost file served from Lambda.
- **Anything not deployed by the video recording time.** A half-working service hurts 'Execution' more than leaving it out.

3.4 Three-minute demo video storyboard

Time	Screen	Say	AWS cut
0:00 to 0:20	Heat map of Delhi (max surface temp 56.4 C, hotspot 374.8 km2)	The problem: heat is unequal. Who needs help first and what should a city build?	-
0:20 to 0:50	Ward Risk (analysis zones), stability chip, Plan for this zone	Risk = heat x people x vulnerability, a relative ranking. 2 km zones until official wards are loaded.	CloudFront URL in the address bar
0:50 to 1:50	Action Plan v2: preset, Rs 50 cr, Build. Headline card, map Before/After, compare sentence, naive benchmark, water panel, roadmap	Estimates are ranges from literature; costs are placeholders. Equity plan protects N more vulnerable people. Optimizer gets Nx more people per rupee than hottest-first.	-
1:50 to 2:20	Generate brief: English and Hindi, PDF, Bedrock badge	Strands agent on Bedrock with guardrails; every number is verified against tool output, template fallback if not.	Bedrock console or CloudWatch BriefSource
2:20 to 2:45	Simulate red alert: email with department tasks, Hindi audio, citizen page	Simulated for the demo, labelled as such. Daily 06:00 IST job in production.	Step Functions graph, SNS email, Polly MP3 in S3
2:45 to 3:00	Model Card and honesty slide	What it is not: air temperature, harm prediction, measured cooling. Next: night layer, ward polygons with census age, post-build measurement.	CloudFormation stack and cost alarm

Part 4. Claude Code handoff

4.1 Paste this at the start of the session

You are extending HeatRelief, an existing, working project (FastAPI backend, React 19 plus Vite frontend, SAM template, 53 passing pytest tests). Rules: (1) Work additively; do not rewrite or rename existing modules, routes or pages, and keep old request shapes working. (2) Run `pytest -q` after every task and keep it green; add tests for new logic. (3) Internal money unit is Rs lakh; 1 crore = 100 lakh. (4) Never invent citations or numbers: placeholders live in config, are named PLACEHOLDER_*, and are labelled on screen. (5) Cooling is always shown as low, mid, high. (6) Use 'analysis zones' unless official ward boundaries are loaded. (7) Every AWS-dependent feature must degrade gracefully to a local or template path. (8) Locate files with grep before editing; file names in this document are guesses. Work through the tasks in order and report each acceptance result.

4.2 Ordered task list

#	Task	Acceptance	Tier
1	Deploy: run the deploy script, fix build and IAM errors, run smoke test against the live URL, add URLs to README (A1, A4).	Live CloudFront URL, smoke passes, README updated.	T0
2	Bedrock live: model access, model comparison on 10 briefs, guardrail test in Hindi, narrowed IAM, UI source badge (A2, F2).	At least 8 of 10 source=bedrock with numbers verified.	T0
3	Reliability: provisioned concurrency, throttle changes, precompute job for sensitivity and rank stability (A3, S4).	Cold request under 10 s; no 429 under 30 parallel calls.	T0
4	UI fixes B1, B2, B9 and B5 (simulate toggle plus Lambda simulate event).	Checks in Part 2 pass; [SIMULATION] banner and subject prefix.	T1
5	Backend for Plan v2: new request params, impact fields, mix, zones, compare with chosen column and headline, naive benchmark, roadmap, assumptions (1.8).	New tests pass; Rs 50 cr equals 5,000 lakh test.	T1
6	Frontend Plan v2 blocks I1 to I7, R1 to R6, R8, R9, R11 and the flow bar (1.2, 1.3).	Acceptance checklist 1.9.	T1
7	Water layer W1 to W3 in optimizer, plan response and water panel; tanker list endpoint and SNS publish.	Water panel shows points, coverage, kl/day, tanker list; placeholders labelled.	T1
8	Cognito (S1), Step Functions PlanLifecycle with SNS fan-out (S2), metrics and tracing (S3).	Anonymous Approve returns 401; execution graph visible; BriefSource metric on dashboard.	T1
9	Record video (Part 3.4) and publish the blog.	Video under 3:00, blog link ready.	T0/T1
10	Only now T2: rank stability chip, public/private column, IMD-aligned thresholds, Translate languages, sensitivity curve, verification plan, citizen boost, whole-plan PDF, Rekognition, Location Service.	One at a time, stop at the deadline.	T2

4.3 Final submission checklist

- Live frontend URL and API URL open from a private browser window; warm-up done; stack stays up until results are announced.
- README: architecture diagram, honesty statement, AWS service list, what is simulated, what is placeholder, live URLs, licence credits.
- Public repo with original history linked; no secrets committed.
- 3-minute video recorded with the console cuts; simulated content is labelled on screen and in speech.
- Blog published on AWS Builder Center and linked.
- Student verification, event check-in and submission form completed before the deadline (confirm the time).
- AWS Budgets alarm active; provisioned concurrency turned off after judging if cost matters.

Caveats. I could not check live AWS pricing, current service limits, the IMD criteria text, or the exact submission clock time, so those are marked 'verify'. The Efficiency-versus-Equity figures quoted come from your screenshot (2,045,959 and 2,880,414 people protected); the 41% difference is simple arithmetic on those two numbers.